Type MS · Maximum or minimum? Using the second derivative

Answer 1 of 8 · Question 7 part c · 2 marks

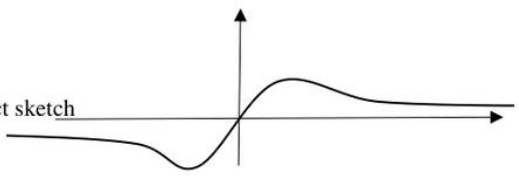
2021 JUN · Paper 01 · extract 1 of 3 · find part c in this mark scheme extract

7 (a)	$\frac{dy}{dx} = \frac{(x^2+4)-x(2x)}{(x^2+4)^2}$	M1
	$= \frac{4-x^2}{(x^2+4)^2}$	A1
	$\frac{dy}{dx} = 0 \Rightarrow 4-x^2 = 0$	M1
	So $\left(2, \frac{1}{4}\right)$ and $\left(-2, -\frac{1}{4}\right)$	A1 A1 (5)
(b)	$\frac{d^2y}{dx^2} = \frac{-2x(x^2+4)^2 - 4x(4-x^2)(x^2+4)}{(x^2+4)^4}$	M1
	$\frac{d^2y}{dx^2} = \frac{-2x^3-8x-16x+4x^3}{(x^2+4)^3}$	M1
	$\frac{d^2y}{dx^2} = \frac{2x^3-24x}{(x^2+4)^3}$	M1
	$\frac{d^2y}{dx^2} = \frac{2x(x^2-12)}{(x^2+4)^3}$ *	A1 cso (4)
(c)	When $x = 2$ $\left[\frac{d^2y}{dx^2} = -\frac{1}{16}\right]$ When $x = -2$ $\left[\frac{d^2y}{dx^2} = \frac{1}{16}\right]$	
	$\frac{d^2y}{dx^2} < 0$ so maximum $\frac{d^2y}{dx^2} > 0$ so minimum	M1 A1ft (2)
		Total 11 marks

Type MS · Maximum or minimum? Using the second derivative

Answer 1 of 8 · Question 7 part c · 2 marks

2021 JUN · Paper 01 · extract 2 of 3 · find part c in this mark scheme extract

	A1	For the correct coordinates of either $\left(2,\frac{1}{4}\right)$ or $\left(-2,-\frac{1}{4}\right)$
	A1	For both correct coordinates $\left(2,\frac{1}{4}\right)$ and $\left(-2,-\frac{1}{4}\right)$
(b)	**M1**	For an attempt at Quotient rule on their $\frac{dy}{dx}=\frac{(x^2+4)-x(2x)}{(x^2+4)^2}$ which must be as a minimum: $\frac{ax^2+bx+c}{(x^2+4)^2}$ where a, b and c are constants and $a,c\neq 0$ The definition of an attempt is that there must be a minimal attempt to differentiate the numerator and denominator and the correct formula applied. $(x^2+4)^2$ must differentiate to $ax(x^2+4)$, the denominator must be $(x^2+4)^4$ Allow the terms in the numerator to be the wrong way around, but the terms must be subtracted. Apply General Guidance for an attempt at differentiation on ax^2+bx+c. $\frac{d^2y}{dx^2}=\frac{-2x(x^2+4)^2-4x(4-x^2)(x^2+4)}{(x^2+4)^4}$
	M1	For cancelling through by (x^2+4) $\frac{d^2y}{dx^2}=\frac{-2x(x^2+4)-4x(4-x^2)}{(x^2+4)^3}$
	M1	For simplifying the numerator to achieve as a minimum $\frac{d^2y}{dx^2}=\frac{ax^3+bx}{(x^2+4)^3}$ where a and b are constants
	A1 cso	For obtaining the answer as given with no errors. $\frac{d^2y}{dx^2}=\frac{2x(x^2-12)}{(x^2+4)^3}$ *
(c)	**M1**	Substitutes either their 2 or their – 2 into their $\frac{d^2y}{dx^2}$ Note: When $x=2\left[\frac{d^2y}{dx^2}=-\frac{1}{16}\right]$ and when $x=-2\left[\frac{d^2y}{dx^2}=\frac{1}{16}\right]$
	A1ft	For the conclusion $\frac{d^2y}{dx^2}<0$ so maximum $\frac{d^2y}{dx^2}>0$ so minimum
	ALT – tests gradient or sight of a sketch	
	M1	Tests gradient on either side of one the turning points (their 2 or their – 2) using their $\frac{dy}{dx}$ or a correct sketch 
	A1ft	For the correct conclusion

Type MS · Maximum or minimum? Using the second derivative

Answer 1 of 8 · Question 7 part c · 2 marks

2021 JUN · Paper 01 · extract 3 of 3 · find part c in this mark scheme extract

Question number	Scheme	Marks

Type MS · Maximum or minimum? Using the second derivative

Answer 2 of 8 · Question 9 part b.i · 0 marks

2019 JUN · Paper 02 · extract 1 of 3 · find part b.i in this mark scheme extract

9(a)	$f(x)=\frac{1}{4}x^4-\frac{1}{3}x^3-2x^2+4x\ (+c)$	M1A1
	$x=-2\ \ y=-\frac{28}{3}\ \Rightarrow c=0$	M1
	$\left(f(x)=\frac{1}{4}x^4-\frac{1}{3}x^3-2x^2+4x\right)\ \therefore C$ passes through O	A1 cso (4)
(b)(i)	$x=2\ \ f'(x)=8-4-8+4=0$	M1
	$\frac{d^2y}{dx^2}=3x^2-2x-4$	M1
	$x=2\quad \frac{d^2y}{dx^2}=12-4-4>0\quad \therefore$ min at $x=2$	A1cso
	$x=1\ \ f'(x)=1-1-4+4=0$	M1
	$x=1\quad \frac{d^2y}{dx^2}=3-2-4<0\quad \therefore$ max at $x=1$	A1 cso
ALT	$f'(x)=(x-2)(x-1)(x+2)\ (=0)$ factorise	M1
	$x=2,1,(-2)$ solve (solutions to be 2,1 (and another))	M1
	OR: $f'(x)\ (=0)$ solved by calculator.	
	All 3 solutions needed (and correct) $=0$ not needed M2	
	$\frac{d^2y}{dx^2}=3x^2-2x-4$ differentiate	M1
	$x=2\quad \frac{d^2y}{dx^2}=12-4-4>0\quad \therefore$ min at $x=2$	A1cso
	$x=1\quad \frac{d^2y}{dx^2}=3-2-4<0\quad \therefore$ max at $x=1$	A1cso
(ii)	$x=1\Rightarrow y=1\frac{11}{12}\qquad x=2\Rightarrow y=1\frac{1}{3}$	B1B1 (7)
(c)	$y'=(x-1)(x-2)(x+2)$	M1
(i)	$x=-2,\ y=-\frac{28}{3}$ or $\left(-2,-\frac{28}{3}\right)$	A1
(ii)	$x=-2\quad \frac{d^2y}{dx^2}=12+4-4>0\quad \therefore$ min point	A1cso (3)
		[14]

Type MS · Maximum or minimum? Using the second derivative

Answer 2 of 8 · Question 9 part b.i · 0 marks

2019 JUN · Paper 02 · extract 2 of 3 · find part b.i in this mark scheme extract

Question Number	Scheme	Marks
(a)M1	Attempt to integrate f'(x). The power of at least one x term must increase and none should decrease. c not needed	
A1	Correct integration, c not needed	
M1	Substitute the given coordinates to show $c = 0$. If c is not included (or assumed to be 0), then showing that substitution of $x = -2$ gives $y = -28/3$ is acceptable. Substitutions must be shown.	
A1cso	Correct conclusion from fully correct work. Accept eg $f(0) = 0$ ∴ shown	
(b)	Ignore labels (i) and (ii) when marking (b)	
(i)M1	Substitute $x = 2$ in the expression for f'(x) to show f'(x) = 0. Substitution must be shown	
M1	Differentiate the expression for f'(x). At least one power must decrease and none increase.	
A1cso	Show second derivative is > 0 at $x = 2$ and give the conclusion. No errors or omissions in the working.	
M1	Substitute $x = 1$ in the expression for f'(x) to show f'(x) = 0. Substitution must be shown	
A1cso	Show second derivative is < 0 at $x = 1$ and give the conclusion. No errors or omissions in the working.	
(ii)B1	For either y coordinate correct (and x coordinate correctly indicated; substitution shown indicates this)	
B1	For the second y coordinate correct	
(c)	(May have been seen in (b))	
M1	Factorise f'(x) completely – any valid method OR use the factor theorem to find $x = -2$	
(i)A1	Extract the x coordinate of the third turning point and obtain the corresponding y coordinate. May quote y coordinate from the question	
(ii)A1cso	Test the sign of the second derivative at this point and make the conclusion. All work in (c) and $\frac{d^2y}{dx^2}$ (from (b)) must be completely correct for this mark to be awarded.	
	Alternative ways to determine the nature of the turning points:	
1.	If the change of sign of f'(x) is used then values of f'(x) either side of 1 and 2 must be calculated to provide evidence.	
2.	The continuity of a cubic function can be used to establish the nature of the turning points. If in doubt send to review.	

Type MS · Maximum or minimum? Using the second derivative

Answer 2 of 8 · Question 9 part b.i · 0 marks

2019 JUN · Paper 02 · extract 3 of 3 · find part b.i in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Maximum or minimum? Using the second derivative

Answer 3 of 8 · Question 9 part c.ii · 3 marks

2019 JUN · Paper 02 · extract 1 of 3 · find part c.ii in this mark scheme extract

9(a)	$f(x)=\frac{1}{4}x^4-\frac{1}{3}x^3-2x^2+4x\ (+c)$	M1A1
	$x=-2\ \ y=-\frac{28}{3}\ \Rightarrow c=0$	M1
	$\left(f(x)=\frac{1}{4}x^4-\frac{1}{3}x^3-2x^2+4x\right)\ \therefore C$ passes through O	A1 cso (4)
(b)(i)	$x=2\ \ f'(x)=8-4-8+4=0$	M1
	$\frac{d^2y}{dx^2}=3x^2-2x-4$	M1
	$x=2\ \ \frac{d^2y}{dx^2}=12-4-4>0\ \ \therefore$ min at $x=2$	A1cso
	$x=1\ \ f'(x)=1-1-4+4=0$	M1
	$x=1\ \ \frac{d^2y}{dx^2}=3-2-4<0\ \ \therefore$ max at $x=1$	A1 cso
ALT	$f'(x)=(x-2)(x-1)(x+2)\ (=0)$ factorise	M1
	$x=2,1,(-2)$ solve (solutions to be 2,1 (and another))	M1
	OR: $f'(x)\ (=0)$ solved by calculator.	
	All 3 solutions needed (and correct) $=0$ not needed M2	
	$\frac{d^2y}{dx^2}=3x^2-2x-4$ differentiate	M1
	$x=2\ \ \frac{d^2y}{dx^2}=12-4-4>0\ \ \therefore$ min at $x=2$	A1cso
	$x=1\ \ \frac{d^2y}{dx^2}=3-2-4<0\ \ \therefore$ max at $x=1$	A1cso
(ii)	$x=1\Rightarrow y=1\frac{11}{12}\qquad x=2\Rightarrow y=1\frac{1}{3}$	B1B1 (7)
(c)	$y'=(x-1)(x-2)(x+2)$	M1
(i)	$x=-2,\ y=-\frac{28}{3}$ or $\left(-2,-\frac{28}{3}\right)$	A1
(ii)	$x=-2\ \ \frac{d^2y}{dx^2}=12+4-4>0\ \ \therefore$ min point	A1cso (3)
		[14]

Type MS · Maximum or minimum? Using the second derivative

Answer 3 of 8 · Question 9 part c.ii · 3 marks

2019 JUN · Paper 02 · extract 2 of 3 · find part c.ii in this mark scheme extract

Question Number	Scheme	Marks
(a)M1	Attempt to integrate f'(x). The power of at least one x term must increase and none should decrease. c not needed	
A1	Correct integration, c not needed	
M1	Substitute the given coordinates to show $c = 0$. If c is not included (or assumed to be 0), then showing that substitution of $x = -2$ gives $y = -28/3$ is acceptable. Substitutions must be shown.	
A1cso	Correct conclusion from fully correct work. Accept eg $f(0) = 0$ ∴ shown	
(b)	Ignore labels (i) and (ii) when marking (b)	
(i)M1	Substitute $x = 2$ in the expression for f'(x) to show f'(x) = 0. Substitution must be shown	
M1	Differentiate the expression for f'(x). At least one power must decrease and none increase.	
A1cso	Show second derivative is > 0 at $x = 2$ and give the conclusion. No errors or omissions in the working.	
M1	Substitute $x = 1$ in the expression for f'(x) to show f'(x) = 0. Substitution must be shown	
A1cso	Show second derivative is < 0 at $x = 1$ and give the conclusion. No errors or omissions in the working.	
(ii)B1	For either y coordinate correct (and x coordinate correctly indicated; substitution shown indicates this)	
B1	For the second y coordinate correct	
(c)	(May have been seen in (b))	
M1	Factorise f'(x) completely – any valid method OR use the factor theorem to find $x = -2$	
(i)A1	Extract the x coordinate of the third turning point and obtain the corresponding y coordinate. May quote y coordinate from the question	
(ii)A1cso	Test the sign of the second derivative at this point and make the conclusion. All work in (c) and $\frac{d^2y}{dx^2}$ (from (b)) must be completely correct for this mark to be awarded.	
	Alternative ways to determine the nature of the turning points:	
1.	If the change of sign of f'(x) is used then values of f'(x) either side of 1 and 2 must be calculated to provide evidence.	
2.	The continuity of a cubic function can be used to establish the nature of the turning points. If in doubt send to review.	

Type MS · Maximum or minimum? Using the second derivative

Answer 3 of 8 · Question 9 part c.ii · 3 marks

2019 JUN · Paper 02 · extract 3 of 3 · find part c.ii in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Maximum or minimum? Using the second derivative

Answer 4 of 8 · Question 7 part b · 2 marks

2023 JUN · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

7(a)	$\frac{dy}{dx} = 2mx + \frac{1}{2} \times 64x^{-\frac{1}{2}}$ $32x^{-\frac{1}{2}} + 2mx = 0 \Rightarrow 32 \times 4^{-\frac{1}{2}} + 2m \times 4 = 0 \Rightarrow 16 + 8m = 0 \Rightarrow m = -2$ $n = 39 + 64\sqrt{4} - 2 \times 4^2 = 135$	M1A1 dM1A1 dM1A1 [6]
(b)	$\frac{d^2y}{dx^2} = -\frac{1}{2} \times 32x^{-\frac{3}{2}} - 4 = -16x^{-\frac{3}{2}} - 4$ $\frac{d^2y}{dx^2} = -6 < 0 \Rightarrow$ maximum	M1 A1FT [2]
		Total 8 marks

Type MS · Maximum or minimum? Using the second derivative

Answer 4 of 8 · Question 7 part b · 2 marks

2023 JUN · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question	Scheme	Marks

Type MS · Maximum or minimum? Using the second derivative

Answer 5 of 8 · Question 7 part a.ii · 3 marks

2022 JUN · Paper 01 · extract 1 of 4 · find part a.ii in this mark scheme extract

7 a (i)	f $x = \int (4x^3 - 12x^2 - 19x + 12)\,dx = \frac{4}{4}x^4 - \frac{12}{3}x^3 - \frac{19}{2}x^2 + 12x + D$ oe For the point $4, -104$ it follows that $-104 = (4)^4 - 4(4)^3 - \frac{19}{2}(4)^2 + 12(4) + D$ * $-104 = 256 - 256 - 152 + 48 + D \Leftrightarrow D = 0$*	M1 A1 M1 A1* cso (4)
a (ii)	$x = 0.5$ $f'(x) = 4(0.5)^3 - 12(0.5)^2 - 19(0.5) + 12 = 0$ $f''(x) = 12x^2 - 24x - 19$ $x = 0.5$ $f''(x) = 12(0.5)^2 - 24(0.5) - 19(= -28) < 0$ Therefore maximum *	M1 M1 A1 cso (3)
b (i)	$f'(x) = (2x-1)(2x^2 - 5x - 12)$ $f'(x) = (2x-1)(2x+3)(x-4) \rightarrow x =$ $x = -\frac{3}{2}$ or $x = 4$, $\left[x = \frac{1}{2}\right]$ $A\left(-\frac{3}{2}, -\frac{333}{16}\right)$ $B(4, -104)$	M1 M1 A1 A1 A1 (5)
b (ii)	$f''(x) = 12(-1.5)^2 - 24(-1.5) - 19(= 44) > 0$ Therefore minimum $x = 4$ $f''(x) = 12(4)^2 - 24(4) - 19(= 77) > 0$ Therefore minimum (Read notes carefully for allocation of these marks) **Alternative** As we have a maximum at $x = 0.5$ and C is a continuous curve $x = -\frac{3}{2}$ is a minimum and $x = 4$ is a minimum	ddM1 A1 A1 (3) {ddM1} {A1}{A1} (3)
		Total 15 marks

Type MS · Maximum or minimum? Using the second derivative

Answer 5 of 8 · Question 7 part a.ii · 3 marks

2022 JUN · Paper 01 · extract 2 of 4 · find part a.ii in this mark scheme extract

Part	Marks	Notes
(a) (i)	**M1**	Attempt to integrate – see general guidance for minimally acceptable attempt – at least one term must be fully correct in this integration (unsimplified).
	A1	Fully correct integration – does not need to include a constant of integration, need not be simplified.
	M1	Substitution of $x = 4$ and $y = -104$ into their expression, which must include a constant of integration.
	A1 cso	Obtains the given result, making it clear the constant of integration is 0 or listing fully the function at the end after fully correct working. There must be no incorrect working for this final mark to be awarded.
Note: candidates may assume $D = 0$ & show, by substituting, (4, -104) lies on the curve **final M1 A1**		
(a) (ii)	**M1**	Substitution of $x = 0.5$ into $f'(x)$ and shows $f'(x) = 0$. A candidate may also solve $f'(x) = 0$ and solve to get $x = 0.5$.
	M1	For finding $f''(x)$ - see general guidance for minimally acceptable attempt at differentiation. At least one term must be fully correct for this differentiation.
	A1 cso*	Substitutes $x = 0.5$ into a fully correct second derivative, shows that the result is negative and draws a conclusion. This can be as simple as "shown" or #. It is not necessary to work out the actual value of $f''(0.5)$ if clear substitution is shown, but if calculated, it must be correct. No incorrect work for this mark to be awarded. Note: as always, this A mark must follow 2 M1 marks.

Type MS · Maximum or minimum? Using the second derivative

Answer 5 of 8 · Question 7 part a.ii · 3 marks

2022 JUN · Paper 01 · extract 3 of 4 · find part a.ii in this mark scheme extract

(b)	Part (i) and (ii) may be marked together	
(i)	**M1**	A clear attempt to divide by $(2x-1)$ or compare coefficients. The student must arrive at a quadratic factor of the form $(2x^2+Ax-12)$ if dividing It is also possible to divide by $\left(x-\frac{1}{2}\right)$ and arrive at a quadratic factor of the form $4x^2+Bx-24$
	M1	Uses a complete method to solve their 3TQ, see general guidance for a minimally acceptable attempt. Must progress to $x=$.
	A1	$x=-\frac{3}{2}$ or $x=4$ Award M1 M1 A1 if **both** correct values are given without working.
	A1	$A\left(-\frac{3}{2},-\frac{333}{16}\right)$ or $B(4,-104)$ Allow x = and y = (clearly paired) for **both** final A marks.
	A1	$A\left(-\frac{3}{2},-\frac{333}{16}\right)$ and $B(4,-104)$ accept 20.8 or better – answer is 20.8125
	Factorising in (a)(ii) and then using in this part can be awarded the marks above.	
(b) (ii)	**ddM1**	Correctly substitutes $x=-\frac{3}{2}$ or $x=4$ into their $f''(x)$. Their values Dependent on both previous M marks in part (b). A fully correct evaluation of the second derivative can imply this mark ie sight of 77 and 44. Part (i) and (ii) may be marked together
	A1	Substitutes $x=-1.5$ **or** $x=4$ into a fully correct second derivative, shows that the result is positive and draws the conclusion this is a minimum. It is not necessary to work out the actual value of $f''(x)$ if clear substitution is shown, but if calculated, it must be correct.
	A1	Substitutes $x=-1.5$ **and** $x=4$ into a fully correct second derivative, shows that the result is positive and draws the conclusion that both values of x give a minimum. It is not necessary to work out the actual value of $f''(x)$ if clear substitution is shown, but if calculated, it must be correct.
ALT	**ddM1**	For stating the curve is continuous and there is a maximum at $x=0.5$ Dep both previous M
	A1	For stating "as we have a maximum at $x=0.5$" and progressing to state either A or B must be a minimum.
	A1	Fully correct conclusion, stating A and B must both be minimum points.
There **MUST** be an appropriate justification for these marks to be awarded, such as that given in M1		

Type MS · Maximum or minimum? Using the second derivative

Answer 5 of 8 · Question 7 part a.ii · 3 marks

2022 JUN · Paper 01 · extract 4 of 4 · find part a.ii in this mark scheme extract

Question number	Scheme	Marks
	4 … 3×500 …	M1 A1

Type MS · Maximum or minimum? Using the second derivative

Answer 6 of 8 · Question 7 part b.ii · 3 marks

2022 JUN · Paper 01 · extract 1 of 4 · find part b.ii in this mark scheme extract

7 a (i)	f $x = \int (4x^3 - 12x^2 - 19x + 12)\,dx = \frac{4}{4}x^4 - \frac{12}{3}x^3 - \frac{19}{2}x^2 + 12x + D$ oe For the point $4, -104$ it follows that $-104 = (4)^4 - 4(4)^3 - \frac{19}{2}(4)^2 + 12(4) + D$ * $-104 = 256 - 256 - 152 + 48 + D \Leftrightarrow D = 0$*	M1 A1 M1 A1* cso (4)
a (ii)	$x = 0.5$ f $'(x) = 4(0.5)^3 - 12(0.5)^2 - 19(0.5) + 12 = 0$ f$''(x) = 12x^2 - 24x - 19$ $x = 0.5$ f$''(x) = 12(0.5)^2 - 24(0.5) - 19 (= -28) < 0$ Therefore maximum *	M1 M1 A1 cso (3)
b (i)	f$'(x) = (2x-1)(2x^2 - 5x - 12)$ f$'(x) = (2x-1)(2x+3)(x-4) \rightarrow x =$ $x = -\frac{3}{2}$ or $x = 4$, $\left[x = \frac{1}{2}\right]$ $A\left(-\frac{3}{2}, -\frac{333}{16}\right)$ $B(4, -104)$	M1 M1 A1 A1 A1 (5)
b (ii)	f$''(x) = 12(-1.5)^2 - 24(-1.5) - 19 (= 44) > 0$ Therefore minimum $x = 4$ f$''(x) = 12(4)^2 - 24(4) - 19 (= 77) > 0$ Therefore minimum (Read notes carefully for allocation of these marks) **Alternative** As we have a maximum at $x = 0.5$ and C is a continuous curve $x = -\frac{3}{2}$ is a minimum and $x = 4$ is a minimum	ddM1 A1 A1 (3) {ddM1} {A1}{A1} (3)
		Total 15 marks

Type MS · Maximum or minimum? Using the second derivative

Answer 6 of 8 · Question 7 part b.ii · 3 marks

2022 JUN · Paper 01 · extract 2 of 4 · find part b.ii in this mark scheme extract

Part	Marks	Notes
(a) (i)	**M1**	Attempt to integrate – see general guidance for minimally acceptable attempt – at least one term must be fully correct in this integration (unsimplified).
	A1	Fully correct integration – does not need to include a constant of integration, need not be simplified.
	M1	Substitution of $x = 4$ and $y = -104$ into their expression, which must include a constant of integration.
	A1 cso	Obtains the given result, making it clear the constant of integration is 0 or listing fully the function at the end after fully correct working. There must be no incorrect working for this final mark to be awarded.
Note: candidates may assume $D = 0$ & show, by substituting, (4, -104) lies on the curve **final M1 A1**		
(a) (ii)	**M1**	Substitution of $x = 0.5$ into $f'(x)$ and shows $f'(x) = 0$. A candidate may also solve $f'(x) = 0$ and solve to get $x = 0.5$.
	M1	For finding $f''(x)$ - see general guidance for minimally acceptable attempt at differentiation. At least one term must be fully correct for this differentiation.
	A1 cso*	Substitutes $x = 0.5$ into a fully correct second derivative, shows that the result is negative and draws a conclusion. This can be as simple as "shown" or #. It is not necessary to work out the actual value of $f''(0.5)$ if clear substitution is shown, but if calculated, it must be correct. No incorrect work for this mark to be awarded. Note: as always, this A mark must follow 2 M1 marks.

Type MS · Maximum or minimum? Using the second derivative

Answer 6 of 8 · Question 7 part b.ii · 3 marks

2022 JUN · Paper 01 · extract 3 of 4 · find part b.ii in this mark scheme extract

(b)	Part (i) and (ii) may be marked together	
(i)	**M1**	A clear attempt to divide by $(2x-1)$ or compare coefficients. The student must arrive at a quadratic factor of the form $(2x^2+Ax-12)$ if dividing It is also possible to divide by $\left(x-\frac{1}{2}\right)$ and arrive at a quadratic factor of the form $4x^2+Bx-24$
	M1	Uses a complete method to solve their 3TQ, see general guidance for a minimally acceptable attempt. Must progress to $x=$.
	A1	$x=-\frac{3}{2}$ or $x=4$ Award M1 M1 A1 if **both** correct values are given without working.
	A1	$A\left(-\frac{3}{2},-\frac{333}{16}\right)$ or $B(4,-104)$ Allow x = and y = (clearly paired) for **both** final A marks.
	A1	$A\left(-\frac{3}{2},-\frac{333}{16}\right)$ and $B(4,-104)$ accept 20.8 or better – answer is 20.8125
	Factorising in (a)(ii) and then using in this part can be awarded the marks above.	
(b) (ii)	**ddM1**	Correctly substitutes $x=-\frac{3}{2}$ or $x=4$ into their $f''(x)$. Their values Dependent on both previous M marks in part (b). A fully correct evaluation of the second derivative can imply this mark ie sight of 77 and 44. Part (i) and (ii) may be marked together
	A1	Substitutes $x=-1.5$ **or** $x=4$ into a fully correct second derivative, shows that the result is positive and draws the conclusion this is a minimum. It is not necessary to work out the actual value of $f''(x)$ if clear substitution is shown, but if calculated, it must be correct.
	A1	Substitutes $x=-1.5$ **and** $x=4$ into a fully correct second derivative, shows that the result is positive and draws the conclusion that both values of x give a minimum. It is not necessary to work out the actual value of $f''(x)$ if clear substitution is shown, but if calculated, it must be correct.
ALT	**ddM1**	For stating the curve is continuous and there is a maximum at $x=0.5$ Dep both previous M
	A1	For stating "as we have a maximum at $x=0.5$" and progressing to state either A or B must be a minimum.
	A1	Fully correct conclusion, stating A and B must both be minimum points.
There **MUST** be an appropriate justification for these marks to be awarded, such as that given in M1		

Type MS · Maximum or minimum? Using the second derivative

Answer 6 of 8 · Question 7 part b.ii · 3 marks

2022 JUN · Paper 01 · extract 4 of 4 · find part b.ii in this mark scheme extract

Question number	Scheme	Marks
	4 … 3×500 …	M1 A1

Type MS · Maximum or minimum? Using the second derivative

Answer 7 of 8 · Question 4 part b · 1 mark

2021 NOV · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

4 (a)	$f(x)=2(x-3)^2-5$ $\Rightarrow f(x)=2(x^2-6x+9)-5=2x^2-12x+13$ $p=-12,*\ \ q=13$ **ALT** $\frac{dy}{dx}=4x+p=0$ at $(3,-5)$ $\Rightarrow 4\times 3+p=0\Rightarrow p=-12*$ $-5=2\times 3^2-12\times 3+q\Rightarrow q=13$	M1M1 M1A1 cso [4] M1 M1 M1A1 cso [4]
(b)	Minimum since coefficient of x^2 is positive	B1 [1]
(c)	$\frac{dy}{dx}=4x-12\Rightarrow$ when $x=1$ $\frac{dy}{dx}=-8$ So gradient of normal $=\frac{1}{8}$ When $x=1$ $y=3$ $y-'3'=\frac{1}{'8'}(x-1)$ Eg. $y-3=\frac{1}{8}(x-1)$, $x-8y+23=0$ oe	B1 B1ft M1 M1 A1, A1 [6]
		Total 11 marks

Type MS · Maximum or minimum? Using the second derivative

Answer 7 of 8 · Question 4 part b · 1 mark

2021 NOV · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
(a)	M1	For using the coordinate of the stationary point and forming the expression: $f(x)=2(x-3)^2-5$ This must be correct for this mark
	M1	Expands their $2(x-3)^2-5$ to form a 3TQ
	M1	For equating coefficients with the given equation to find the value of p and q
	A1	For $p=-12,*\ q=13$ **NB:** The value of p is given in the question.
	ALT – Uses differentiation	
	M1	For a correct differentiation of the expression for y This must be correct for this mark
	M1	For equating their $\frac{dy}{dx}=0$, substituting in $x = 3$ and attempting to find a value for p
	M1	For substituting the given value of p into $y=2x^2+px+q$ using the coordinates $(3,-5)$ to find the value of q. Explicit substitution of the value of p into the equation must be seen if p has not been found using differentiation. [M0M0M1A0 can be scored]
	A1	For $p=-12,*\ q=13$ **NB:** The value of p is given in the question.
(b)	B1	For the correct statement: Minimum since coefficient of x^2 is positive
	ALT	
	B1	For $\frac{d^2y}{dx^2}=4\Rightarrow$ positive, hence minimum. A correct conclusion is required.
(c)	B1	For the value of the gradient when $x = 1$ $[m=-8]$
	B1ft	For the gradient of the normal which must have come from differentiation. Ft their -8
	M1	For the value of y when $x = 1$ using their value of q
	M1	For a complete equation of the line in any form using their normal and their value of y Allow this mark for an equation of the line with $x = 1$ and $y =$ '3' with their normal, provided there is evidence that the normal comes from the negative reciprocal of their value of the tangent. For this mark the gradient of the tangent need not have come from differentiation.
	A1	For the correct equation in any form.
	A1	For the correct equation in the required form or any multiples of the coefficients.

Type MS · Maximum or minimum? Using the second derivative

Answer 7 of 8 · Question 4 part b · 1 mark

2021 NOV · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Maximum or minimum? Using the second derivative

Answer 8 of 8 · Question 4 part b · 2 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part b in this mark scheme extract

4 (a)	$\frac{dy}{dx} = 3x^2 - 6x - 24$ $"(x+2)(x-4)" = 0$ $x = -2$ or $x = 4$ (–2, 34) and (4, –74)	M1 M1 A1 A1 (4)
(b)	$\frac{d^2y}{dx^2} = "6x - 6"$ and substitution of their –2 or 4 or consider values of $\frac{dy}{dx}$ either side of their –2 or 4 or use properties of a cubic* (–2, 34) maximum and (4, –74) minimum	M1 A1cso (2) **[6]**

Part	Mark	Additional Guidance
(a)	M1	General guidance – an attempt to differentiate, power of at least one term must decrease by one. Also, no power must increase.
	M1	Equates their derivative = 0 and attempts to solve by any method – **see general guidance.**
	A1	Both correct values for x
	A1	All values correct, listed as coordinates or correctly paired $x = \ldots\ldots$ $y = \ldots\ldots$
(b)	M1	**Correctly differentiates their derivative** from part a and substitutes one of their x values. Allow sight of 18 or -18 following a correct differentiation to imply this mark.
	A1 cso	Allow this mark if there is an incorrect y value from part a Must correctly make the argument they've chosen and identify the points as shown. If substituting x = -2 and x = 4 or values either side of the these, their evaluations of the substitutions must be correct. * A convincing argument about the shape of a positive cubic curve and position of the minimum and maximum point must be made. Send to review, if in doubt. In the conclusion, it must be clearly stated which coordinate or value of x is a maximum and which is a minimum. Just a substitution of a value and stated max or min is insufficient.

Type MS · Maximum or minimum? Using the second derivative

Answer 8 of 8 · Question 4 part b · 2 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part b in this mark scheme extract

Question number	Scheme	Marks